Carrier sizing correction and closed-loop N+1 simulation

23 October 2026 | MATLAB/Simulink R2026a | Engineering simulation, not commissioned equipment

Seven executed annual simulation scenarios

A new versioned `DC_Pool_Capacity_Fault_20261023_SK.slx` includes a time-dependent district heat pump output cap at **every 300-second step**. The water temperature controller, thermal storage and backup gas are dynamically recalculated, instead of clipping prerecorded output.

Scenario	Annual district gas MWh
No pool, 12 MW HP limit	9085.895
No pool, 10 MW HP limit	9088.579
Pool, 10 MW HP limit	9191.507
Pool, 24 hours with one 2-MW module unavailable	9207.071
Pool, N+1 replacement available after 30 minutes	9191.637
Pool, HP output limited to 9.42 MW	9256.806
Pool, HP output limited to 10.99 MW	9189.267

All seven scenarios ran for 8760 hours and 105120 steps. Thermal balance, source exclusivity and the modeled pool water ceiling of 27.8 °C passed. Peak pool temperature: **27.62 °C**.

Manufacturer correction

The **Carrier 61XWH-ZE 2025 product data** reports a maximum **1570 kW** nominal heat output for model 17. Its documented test conditions are evaporator entering/leaving water **20/15 °C**, condenser **70/75 °C** according to EN14511. This is NOT a validated 35/30 °C waste-heat source to 50/70 °C district heating rating.

- 6 operating model 17 units: 9.42 MW nominal + 1 spare: **7 total**.
- 7 operating model 17 units: 10.99 MW nominal + 1 spare: **8 total**.
- The Simulink difference is 67.538 MWh/year of backup gas and 14.145 MWh/year higher electricity demand with the larger bank.
- At selected Eurostat Slovakia 2025 electricity/gas prices, the bigger bank saves **€2,023.92/year** in modeled variable energy expenditure.

Investment implication and boundaries

With 2% of extra CAPEX as annual fixed maintenance, a 6% discount rate and 20-year model lifetime, the break-even incremental investment for the seventh operating (eighth installed) unit is only **about €18,883**. That is a screening threshold, not a supplier quote, and does not price reliability or standby service.

Critical missing evidence: Manufacturer selection report for the exact 35/30 °C source and 50/70 °C network (also 25/20 °C source), including full/part-load thermal output, COP, pressure losses, minimum turndown, start current and N+1 recovery time. The model's 30-minute failover is a scenario assumption, not real hardware certification.

IT waste-heat reuse does **not** replace independent critical cooling. Size independently to reject approximately **10 MW of rated IT heat** under full IT demand, with redundancy, electrical autonomy and safe failure modes established in engineering design.

References: Carrier 2025 catalogue, physical ratings on PDF page 8, indicative single/multi-unit COP plot on page 9; Eurovent Certified Carrier model 07.

Reproducible data: `capacity_fault_simulink_2025.csv`, `capacity_fault_simulink_2025.json`, `Carrier_2025_nominal_vs_Nplus1_options.csv`, `Carrier_2025_rated_vs_simulated_economics.json`.